

Exoplanets by Host Star Spectral Type: Linked Views of Radius and Equilibrium Temperature

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1. Introduction

This project develops an interactive visualization system (web application) to explore how exoplanet properties differ by the host star's spectral type. The research question is: **How do the distributions of exoplanet radius and equilibrium temperature vary across F/G/K/M-type host stars, and what patterns become visible through linked interactive views?** The system is implemented as a GitHub Pages site using HTML/CSS/JavaScript and D3 (v6), and it uses both a preprocessed static dataset stored in the repository and an API-based “live” dataset retrieval for updates.

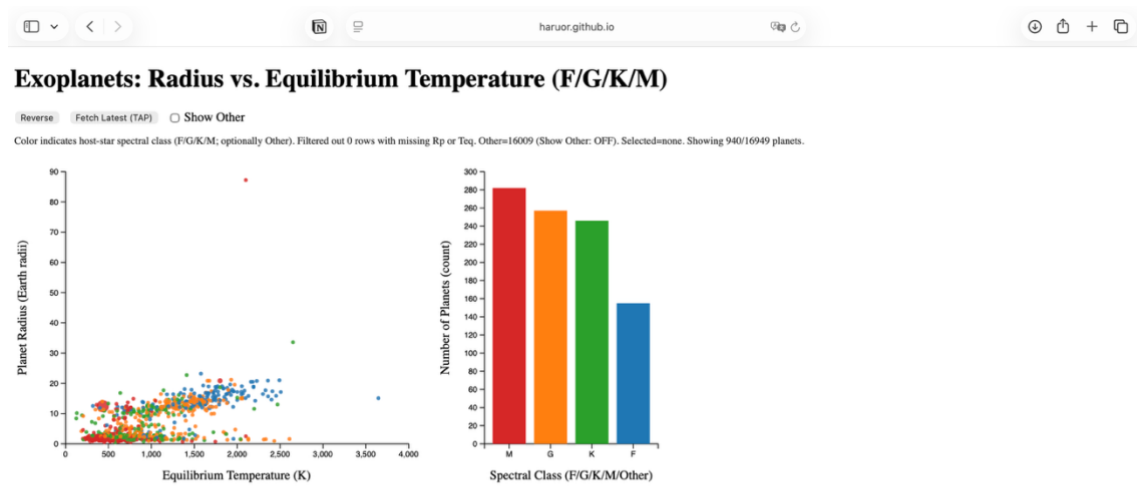


Fig.1: Snapshot of the visualization system (scatter plot + bar chart with controls). Replace this box with a screenshot.

2. Method

Dataset. The static dataset (exoplanets_static.csv) is derived from NASA's Exoplanet Archive Planetary Systems (PS) data. For visualization, the system uses the following columns: planet name (pl_name), planet radius in Earth radii (pl_rade), equilibrium temperature in Kelvin (pl_eqt), orbital period (pl_orbper), and host star spectral type (st_spectype). The “Fetch Latest (TAP)” function queries the Exoplanet Archive TAP endpoint via a CORS-enabled proxy and updates the in-memory dataset.

Views. (1) A **scatter plot** shows the relationship between equilibrium temperature (x-axis) and radius (y-axis);

each point is an exoplanet. (2) A **bar chart** summarizes the number of planets by spectral type category. The system focuses on the four major classes (F, G, K, M). An optional “Show Other” mode aggregates non-F/G/K/M spectral types into an “Other” category.

Interactions and linking. The design follows the course examples for (i) animated bar transitions (reverse sorting) and (ii) tooltips for point inspection, and it implements linked interactions between the bar chart and scatter plot by filtering/highlighting selected categories. Clicking a bar toggles a filter state; the scatter plot updates to show the selected categories. To reduce reliance on a separate legend, the color assignment is kept consistent across both views so that color works as an implicit cross-view key.

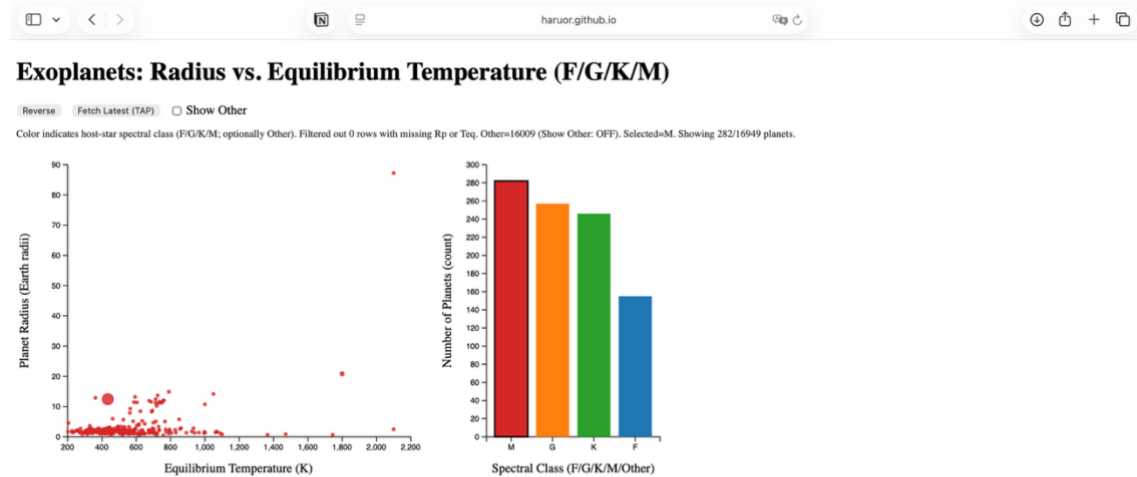


Fig.2: Example interaction: selecting one or more spectral types on the bar chart and observing the linked update in the scatter plot.

3. Result

The implemented system enables interactive exploration of the dataset through coordinated multiple views. Users can (i) obtain an overview of category sizes with the bar chart, (ii) inspect planet-level details with the scatter plot, and (iii) narrow the analysis by selecting spectral categories. The “Reverse” control changes the bar order with an animated transition, improving readability when comparing categories under different sort directions. The tooltip interaction supports rapid inspection of individual planets (name and quantitative attributes) without cluttering the plot. The “Fetch Latest (TAP)” function demonstrates API-based data access by retrieving updated records from the Exoplanet Archive and refreshing the views.

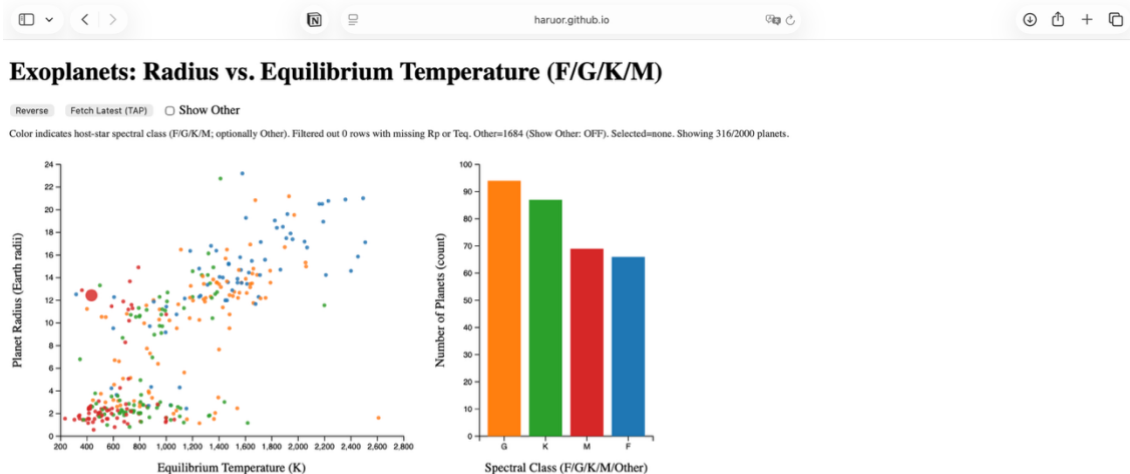


Fig.3: Example of the latest-data retrieval (TAP) and the updated visualization state after refreshing.

4. Discussion

A key design choice is the use of **consistent color mapping** across both the bar chart and scatter plot, which effectively plays the role of a legend while also strengthening the linked-view concept. The “Show Other” option provides context without overwhelming the main comparison among F/G/K/M classes. A limitation is that the analysis depends on the availability and completeness of the chosen columns (e.g., missing radius or equilibrium temperature values are excluded). In addition, the live query is constrained by browser access control (CORS), so a proxy layer is required for client-side retrieval from GitHub Pages.

5. Conclusion

This project delivers a D3-based interactive visualization system that combines a scatter plot and a bar chart to explore exoplanet properties by host-star spectral type. By integrating coordinated interactions (filtering, tooltips, and animated transitions) and supporting both static and API-based datasets, the system satisfies the final-task requirements for multiple linked views and effective interaction design.

6. Reference

- [1] Course handouts and examples (W06, W10, W12) and Final Task description (InfoVis 2025).
- [2] NASA Exoplanet Archive documentation (TAP and Planetary Systems table/columns).
- [3] Cloudflare Workers documentation (CORS header proxy example).
- [4] D3.js v6 documentation.

Submission URLs (replace with your actual URLs):

System: <https://your-account.github.io/InfoVis2025/FinalTask/system.html>

Report: <https://your-account.github.io/InfoVis2025/FinalTask/report.pdf>

Thumbnail: <https://your-account.github.io/InfoVis2025/FinalTask/thumbnail.png>